25901819 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCZIM102 - MOLECULES OF LIFE

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. Carbohydrates are also known as ..... [U] / [CO1]
2. Myoglobin contains a ……. group, which help in oxygen binding. [U] / [CO3]
3. .....is the principal storaage form of energy in adipose tissues. [K] / [CO4]
4. Phospholipid have a.....head and two non polar tail. [U] / [CO5]
5. Galactose is the.....epimer of glucose [U] / [CO1]
6. …….is a polysaccharide serve as supporting tissues of plants. [K] / [CO1]
7. The angle of rotation in peptide bond is known as.....plot. [U] / [CO2]
8. .....can be defined as esters of fatty acid with alcohol. [U] / [CO4]

9. A change in single amino acid in the primary structure due to a mutation can lead to disease known as..... [K] / [CO3]

10. .....are simple proteins having high affinity to sugar groups [U] / [CO2]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Describe the importance of chitin. [K] / [CO1]

12. What is decarboxylation reaction in amino acids? [K] / [CO2]

13. Define lipoproteins. [U] / [CO5]

14. Write short note on lipoprotein. [U] / [CO2]

15. Outline chemical structure of nucleic acids. [U] / [CO5]

16. List out the the main types of disaccharides found in nature. [K] / [CO1]

17. Name two factors that can cause protein denaturation. [U] / [CO2]

18. Explain the relationship between iodine value and double bond. [U] / [CO4]

19. Write a note on formation of Osazone with phenyl hydrazine. [U] / [CO1]

20. What are the complementary base pairs in DNA according to Watson and Crick. [K] / [CO5]

21. Distinguish between amylose aand amylopectin. [U] / [CO1]

22. Write any two properties of zwitter ion [K] / [CO2]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Write an essay on denaturation and renaturation of DNA. [K] / [CO5]

24. Differniate between epimer and anomer. [U] / [CO1]

25. Illustrate the structure of acidic amino acids. [A] / [CO2]

26. Compare and contrast fructose with glucose including their structure. [U] / [CO1]

27. Define the quaternary structure of proteins and explain how it differ from tertiary structure. [K] / [CO3]

28. compare and contrast saturated fatty acids from unsaturated fatty acids. [U] / [CO5]

29. Discuss the significance of heparin. [K] / [CO1]

[4x5 = 20]